

Goal. Explore more methods of determining whether series converge or diverge.

1. State the integral test in your own words. What three conditions do you need to check in order to use the integral test?
2. Use either the n th term divergence test or the integral test to determine whether the following series converge or diverge. Be sure to verify the hypotheses, state your conclusion, and say which test you used.

(a) $\sum_{n=1}^{\infty} (1.01)^n$

(b) $\sum_{n=1}^{\infty} e^{-3n}$

3. Use the integral test to determine for which values of p the following series converges.

$$\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^p}$$

4. For which integer values of $a > 0$ does the following series converge? Explain.

$$\sum_{n=a}^{\infty} \frac{1}{n}$$

5. For which values of p does the following series converge? Explain.

$$\sum_{n=5}^{\infty} \frac{5}{n^{2p+9}}$$